Experiment-6

Write a python program for Vacuum Cleaner problem

Aim

To write and execute a Python program to implement the Vacuum Cleaner Problem using a simple reflex agent that cleans two rooms, A and B.

Algorithm

1. Start with two rooms, A and B, and place the vacuum cleaner in Room A.
2. Represent the status of each room as 1 (Dirty) or 0 (Clean).
3. Set the initial status of both rooms as Dirty.
4. Check the status of the current room.
5. Perform the following actions:
    - If the room is Dirty, suck the dirt and mark the room as Clean.
    - If the room is Clean, do nothing.
    - Move the vacuum cleaner to the other room.
6. Add 1 to the cost every time dirt is cleaned.
7. Repeat the steps until both rooms have been checked.
8. Display the final condition of the rooms and the total cleaning cost.

Coding

ex-6.py - D:/ex-6.py (3.10.0)

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```
# Vacuum Cleaner Problem (Simple Reflex Agent)
# Two rooms A and B  ->  1 = Dirty , 0 = Clean

status = {"A": 1, "B": 1}
location = "A"
cost = 0
visited = 0

print("VACUUM CLEANER PROBLEM")
print("----------------------")
print("Initial Condition :", status)
print("Vacuum is placed in Room", location)
print()

while visited < 2:

    # Percept : check the current room
    if status[location] == 1:
        print("Room", location, "is Dirty -> Suck")
        status[location] = 0
        cost = cost + 1
        print("Room", location, "is now Clean")
    else:
        print("Room", location, "is already Clean")

    visited = visited + 1

    # Move to the other room
    if visited < 2:
        if location == "A":
            new = "B"
        else:
            new = "A"
        print("Moving from Room", location, "to Room", new)
        location = new
    print()

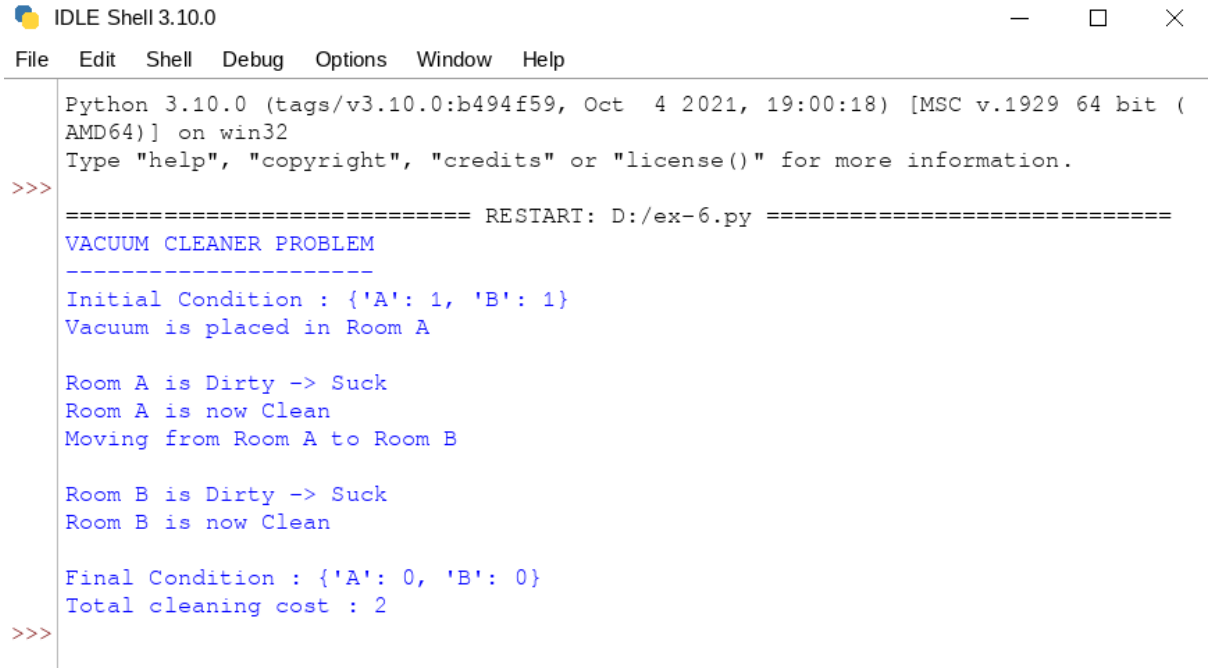
print("Final Condition :", status)
```

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```
print("Total cleaning cost :", cost)
```

Output



Result

The Vacuum Cleaner Problem was successfully solved using a simple reflex agent. The program cleaned both Room A and Room B with a total cleaning cost of 2.